NATIONAL STUDENT IDEATION CHALLENGE / CONCEPT DOCUMENT

JanDrishti

Public Works Verification & Accountability Platform

Connecting public project commitments to on-ground evidence.

CORE PROPOSITION

Make every public works project traceable from what was promised to what was reported, independently observed and ultimately delivered.

SUBMISSION PROFILE	DETAIL
Theme	Surakshit Bharat – Digital Public Infrastructure
Proposed track	Track A – Product or Technology
Product type	Web platform + mobile-friendly field verification interface
Initial context	Urban local body / municipal public works
Current stage	Idea – concept and requirements documented

A proposed accountability layer that complements existing government project-management and procurement systems rather than replacing them.

01 / Executive Summary

JanDrishti is a proposed digital platform for creating an evidence-backed, traceable record of publicly funded infrastructure projects. It connects the original project commitment, contractor-reported progress, independent field observations and subsequent corrective action.

Public project information is often distributed across work orders, progress reports, inspection records and departmental systems. This fragmentation can make it difficult for citizens and reviewers to establish whether reported completion matches approved specifications and actual site conditions. JanDrishti addresses this reconciliation gap.

The platform gives authorities a structured way to register projects, define milestones, collect evidence, conduct independent inspections, identify potential discrepancies and track resolution. Citizens can view information approved for public disclosure and submit observations for review.

The initial MVP is deliberately narrow: municipal road repair and resurfacing within a selected ward. This category offers observable worksites, mappable locations, measurable quantities and understandable before-and-after evidence.

CORE PROPOSITION

Make every public works project traceable from what was promised to what was reported, independently observed and ultimately delivered.

Product principles

- **Evidence before claims:** completion status should be supported by relevant evidence.
- **Independent verification:** contractor submissions and independent observations remain distinguishable.
- **Traceability:** important updates retain their source, timestamp and history.
- **Explainability:** a discrepancy shows what does not match and why it was flagged.
- **Human accountability:** software flags inconsistencies; authorized people make official decisions.
- **Citizen accessibility:** public information is presented in an understandable form.

02 / Problem and Opportunity

The core issue is not only whether project information exists, but whether separate records can be reconciled with independently observed outcomes.

Public infrastructure delivery involves departments that approve projects, contractors that execute them, field officers that inspect them, authorities that certify progress and citizens who depend on the resulting assets. Records generated by these stakeholders may be held in different documents and systems.

This fragmentation creates practical gaps:

• Citizens cannot easily compare the original commitment with actual delivery.

• A completion submission may not establish whether work meets approved specifications.

• Contractor-reported progress and field inspection findings may be difficult to reconcile.

• Defects discovered after completion may not remain connected to the original project record.

• Responsibility and follow-up for discrepancies may be unclear.

• Information may be technically available but difficult for non-specialists to interpret.

JanDrishti's proposed opportunity

Create one project-level record that connects what was approved, what was reported, what was independently observed and how any mismatch was resolved. The platform is designed as an accountability layer alongside existing systems, not as a replacement for procurement, payment or grievance platforms.

Target users and responsibilities

ROLE	NEED / RESPONSIBILITY
Government project administrator	Register projects, approve specifications, assign roles and monitor delivery.
Contractor / implementing agency	Submit milestone progress, quantities and supporting evidence; respond to discrepancies.
Field inspector	Record independent observations, site evidence, quantities and defects.
Independent verifier / auditor	Review evidence, assess discrepancies and record findings.
Citizen	View public project information and submit observations.
Platform administrator	Manage permissions, configuration and audit records.

03 / Proposed Solution

A lifecycle record that links project commitments to execution evidence, independent verification and resolution.

A. Public Project Registry

Each project receives a unique JanDrishti Project ID and a central record containing title, department, implementing agency, location, approved budget, work-order reference, planned dates, specifications, milestones and public documents.

B. Project Lifecycle Tracker

Projects move through defined states: proposed, approved, work awarded, work started, in progress, awaiting inspection, verified, completed, under maintenance and closed. Exception states include delayed, disputed, verification failed, defect identified and rework required.

C. Milestone and Evidence Management

Contractors submit photographs, videos, submission dates, reported completion percentages, quantities, supporting documents and explanations of deviations. Evidence is linked to the relevant project and milestone.

D. Independent Field Verification

Inspectors use a mobile-friendly interface to capture GPS coordinates and reported accuracy, device timestamp, server receipt time, photos or videos, checklist results, observed quantities, defects and verification status. Offline capture and later synchronization are planned.

E. Evidence Reconciliation

The system compares approved commitments, contractor-reported progress and independent field observations. Rule-based checks identify missing evidence, location mismatches, duplicate files, milestone-sequence anomalies and quantity differences. Computer vision may assist with image similarity and before/after comparison.

F. Discrepancy Resolution

Potential mismatches become issue records with a unique ID, linked project and milestone, evidence references, responsible role, response deadline, response history and resolution status. Closure requires a recorded finding and supporting evidence.

G. Citizen Transparency Portal

Citizens search by district, local body, ward, category, status or location. Public pages show permitted project details, milestone progress, verification results, unresolved issues and maintenance history. Citizen reports remain unverified until reviewed.

Lifecycle of evidence and accountability

Each transition retains its timestamp, responsible actor and supporting record.

04 / How the Platform Works

The MVP demonstrates one complete accountability loop rather than attempting to digitize every public works process.

STEP	ACTIVITY	RECORD / RESULT
1. Register	Administrator records approved project scope, budget, location, work order and milestones.	Unique project record and baseline commitments.
2. Report	Contractor submits progress, quantities, photographs and supporting documents.	Pending-verification milestone submission.
3. Validate	System checks required fields, file hashes, milestone sequence, quantities and location metadata.	Validation results and potential alerts.
4. Inspect	Inspector visits the site and captures observations, photographs, location and measured condition.	Independent inspection record.
5. Reconcile	System compares approved, reported and observed information.	Explainable potential discrepancy.
6. Resolve	Responsible party responds; corrective work and follow-up inspection are recorded where required.	Issue history and evidence-backed resolution.
7. Publish	Permitted information is displayed to citizens.	Readable project history and current verification status.

Integrity safeguards

• A contractor's completion submission must never automatically become independently verified completion.

• GPS coordinates and device timestamps are evidence signals, not proof of authenticity by themselves.

• Original evidence uploads should be retained; annotated or processed versions must remain distinguishable.

• Automated alerts must explain the mismatch and required follow-up; they must not label a person or organization corrupt.

• Administrative response and physical resolution must be represented as separate outcomes.

Illustrative demonstration scenario

A hypothetical municipality registers a road repair project to resurface 500 metres of road with an approved budget of ■8 lakh. The contractor reports all milestones complete and submits evidence for 500 metres. An independent inspection records approximately 320 metres resurfaced. JanDrishti flags the reported-versus-observed quantity mismatch, creates a discrepancy record and keeps the project pending verification or disputed. After an explanation and any required corrective work, an inspector records a follow-up finding. The citizen page displays the commitment, reported completion, inspection status and resolution.

This is a designed demonstration scenario, not a real-world test result.

05 / Innovation and Differentiation

The proposed innovation is the integration of separate evidence streams into a consistent commitment-to-verification-to-resolution workflow.

CAPABILITY	JANDrishti APPROACH
Commitment-to-outcome traceability	Connect approved specifications and milestone deliverables to later submissions and observations.
Evidence reconciliation	Compare contractor evidence with independent field observations instead of displaying reported progress alone.
Explainable discrepancy detection	Show expected condition, submitted information, observed condition, reason for alert and required verification.
Discrepancy-centred accountability	Assign each mismatch an owner, deadline, response history and resolution record.
Longitudinal verification	Retain post-completion defects, corrective work and maintenance history.
Citizen-readable transparency	Present project commitments, reported status, verification findings and unresolved issues in accessible language.

What JanDrishti does not claim

The concept does not claim to invent geotagging, digital procurement, project tracking or grievance reporting. It does not replace GeM, CPGRAMS or municipal e-governance platforms, authorize or withhold payments, authenticate every government document, or automatically determine corruption or fraud. Its proposed role is to complement existing systems by reconciling evidence and preserving the history of verification and resolution.

Technology-assisted checks

• **Rule-based validation:** location outside permitted boundary, missing mandatory evidence, inconsistent milestone sequence, duplicate file hashes, unusually short intervals between stages and reported quantity exceeding approved quantity.

• **Computer vision assistance:** visually similar image detection, before-and-after comparison and highlighting potential visible defects for inspector review.

• **Human review:** authorized inspectors and officials make formal verification and resolution decisions.

06 / Proposed Technical Architecture

A modular web application with geospatial project records and protected evidence storage.

USER INTERFACES
Citizen • Contractor • Field Inspector • Administrator
APPLICATION
Responsive React web app / mobile-friendly Progressive Web App
CORE SERVICES
Authentication • Role-based access • Project / milestone / inspection / issue workflows
DATA AND INTELLIGENCE
PostgreSQL + PostGIS • Supabase Storage • Validation services / image checks

Proposed technology stack

LAYER	PROPOSED TECHNOLOGY	RESPONSIBILITY
Frontend	React.js, Tailwind CSS	Responsive web application and mobile-friendly PWA.
Mapping	MapLibre or Leaflet	Display project locations and support location-based review.
Backend and data	Supabase, PostgreSQL, PostGIS	Project records, spatial queries, authentication and application data.
Evidence storage	Supabase Storage	Store original photographs, videos and supporting files.
Access control	Supabase Auth and Row Level Security	Authentication and role-based data access.
Evidence intelligence	Python / FastAPI if needed, OpenCV, perceptual hashing	Image comparison, similarity checks and validation services.
Deployment	Vercel and Supabase	Frontend hosting and managed backend services.

Core data entities

Project: project ID, title, department, implementing agency, contractor, category, budget, dates, location geometry, work-order reference and status.
Milestone: milestone ID, project ID, name, approved specification, expected and reported quantity, planned date, reported status and verification status.
Evidence: evidence ID, project and milestone IDs, submitter, file path, original hash, captured and received timestamps, coordinates, location accuracy, source and verification status.
Inspection: inspection ID, project and milestone IDs, inspector, date, observed condition and quantity, location, findings and decision.
Discrepancy: issue ID, project and milestone IDs, type, description, evidence references, severity, assignee, status and resolution details.
Activity Log: actor, action, previous and new values, timestamp.

- Server-side authorization checks are required for protected operations.
- Evidence files should be hashed and original uploads preserved.
- Status changes and important actions should be recorded in an audit history.
- The system should support retries and idempotent synchronization to prevent duplicate records.
- Only information authorized for public disclosure should appear on citizen-facing pages.

07 / MVP Scope and Implementation Roadmap

Start with one observable infrastructure category and validate the full workflow before expanding.

Recommended pilot: municipal road repair and resurfacing within one selected ward. The category is physically observable, mappable and suitable for milestone-based evidence and quantity checks.

PRIORITY	FEATURES
Must have	Role-based login; project registry; project map; milestone creation; contractor evidence submission; field inspection interface; location and timestamp recording; duplicate-image detection; rule-based discrepancy detection; discrepancy resolution; citizen project page; audit history.
Should have	Tamil and English interface; offline evidence capture and later synchronization; before-and-after image comparison; budget and milestone visualization; citizen observation submission.
Could have	AI-assisted visual defect detection; OCR for tender and work-order documents; cross-project image similarity; automated extraction of project specifications; authorized external data integration.

Implementation phases

PHASE	ACTIVITIES	OUTPUT
1. Problem validation	Interview municipal engineers, contractors, inspectors and citizens; examine work orders and completion reports; map existing inspection processes.	Validated workflow assumptions and pilot requirements.
2. Prototype	Select one ward; create 5–10 sample project records; implement contractor submission, independent inspection and controlled discrepancy cases.	Functional prototype demonstrating the accountability loop.
3. Field validation	Obtain permission to observe or inspect selected worksites; compare records with observations; measure evidence quality and detection performance.	Pilot findings, user feedback and measured validation results.
4. Deployment pathway	Demonstrate potential data exchange with an existing municipal system; document authorization and data-sharing requirements.	Institutional pilot proposal and integration pathway.

Explicitly outside the MVP

Replacing procurement systems; replacing GeM, CPGRAMS or municipal e-governance platforms; authorizing or withholding payments; automatically declaring corruption or fraud; national-scale deployment; facial recognition or Aadhaar processing; and claiming GPS metadata alone proves genuine site capture.

08 / Impact, Measurement and Feasibility

Expected beneficiaries and impact

BENEFICIARY	EXPECTED BENEFIT
Citizens	Accessible information about project commitments, reported progress, verification findings and unresolved issues.
Municipal authorities	Consolidated project records and a structured workflow for identifying and resolving discrepancies.
Inspectors and auditors	Standardized field observations and organized evidence linked to project milestones.
Contractors and agencies	Clear milestone requirements, structured submissions and transparent responses to discrepancies.
Communities	Greater visibility into infrastructure delivery, defects and corrective actions.

Success metrics

- Percentage of projects with complete milestone records.
- Percentage of reported milestones independently inspected.
- Percentage of evidence submissions passing validation.
- Number of discrepancies identified and reviewed.
- Average time between discrepancy identification and response.
- Percentage of discrepancies closed with supporting resolution evidence.
- Precision of duplicate-image detection on a labelled test set.
- Agreement between independent inspectors and system-generated alerts.
- Number of false alerts generated during controlled testing.

Targets should be established after collecting baseline data. No improvement percentages are claimed before a pilot.

Indicative MVP cost

COST HEAD	ESTIMATED RANGE
Frontend development and deployment	■2,000–■5,000
Database, authentication and cloud storage	■0–■5,000
Domain and deployment-related expenses	■1,000–■2,000
Evidence processing and image-validation infrastructure	■3,000–■8,000
Testing, devices, connectivity and demonstration	■4,000–■8,000
Contingency and miscellaneous	■10,000–■12,000
Estimated initial MVP total	■20,000–■40,000

Estimate assumptions: student-team development, open-source technologies, existing laptops and limited sample data. Developer salaries, large-scale infrastructure, government integration and extensive field operations are excluded. A real municipal pilot requires a separate budget.

09 / Risks, Safeguards and Governance

RISK	MITIGATION
Fake GPS or manipulated timestamps	Treat metadata as a signal, not proof; combine with independent inspection, server receipt records and human review.
Reused photographs	Use cryptographic file hashes and perceptual similarity checks; preserve originals.
Poor image quality	Provide capture guidance and require inspector review.
Incorrect automated alerts	Make alerts explainable and require human approval for official decisions.
Lack of government API access	Start with manually uploaded project records and sample work orders.
Citizen misuse	Moderate submissions and verify them before public display.
Privacy leakage	Use role-based visibility, controlled publication and retention policies.
Overlap with existing systems	Position JanDrishti as an evidence-reconciliation and verification layer.

Privacy and responsible use

The platform should publish only information authorized for public disclosure. Private phone numbers, personal identity documents and sensitive location information should not be exposed. Internal inspection records must be restricted to authorized users. Retention and deletion policies should be defined for submitted evidence. The proposed MVP excludes facial recognition and Aadhaar processing.

Potential adopters and implementation partners

Potential adopters include urban local bodies, municipal corporations, municipal engineering and public works departments, relevant state-level urban development bodies, and civic-technology or academic partners. In Tamil Nadu, the Greater Chennai Corporation and other municipal corporations are potential stakeholders for exploratory discussions. These are prospective adopters, not confirmed partners.

The first institutional step is to consult municipal engineering professionals, understand existing inspection and certification workflows, and seek permission for a narrowly scoped pilot. Any integration with government systems must use authorized APIs or approved data-sharing mechanisms.

Current stage and evidence

JanDrishti is currently at the idea and requirements-definition stage. The concept is supported by a documented PRD covering stakeholders, modules, workflows, functional and non-functional requirements, architecture, data entities, MVP scope, metrics, risks and a proposed validation plan. The 500-metre versus 320-metre road resurfacing example is a hypothetical demonstration scenario, not a real-world test. A functional prototype and field validation are future milestones.

JANDrishti

“JanDrishti connects public project promises to on-ground evidence, helping authorities and citizens verify whether taxpayer-funded infrastructure was actually delivered.”

Document status: Proposed concept | Version 1.0 | Prepared for the National Student Ideation Challenge